

EN401: Energy Systems Modelling

Visual Atlas of Model Inputs & Scenario Results (HO3 – HO6)

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Modelling Framework: OSeMOSYS GNU MathProg	Planning Horizon: 2021 – 2035 (15 Years)

PART 1: INPUT PARAMETERS VISUALIZED

The model is driven by exogenous energy demand projections, technology cost assumptions, and physical operational lifetimes calibrated across the 2021–2035 planning horizon.

1.1 Exogenous Final Electricity Demand Growth (ELC003)

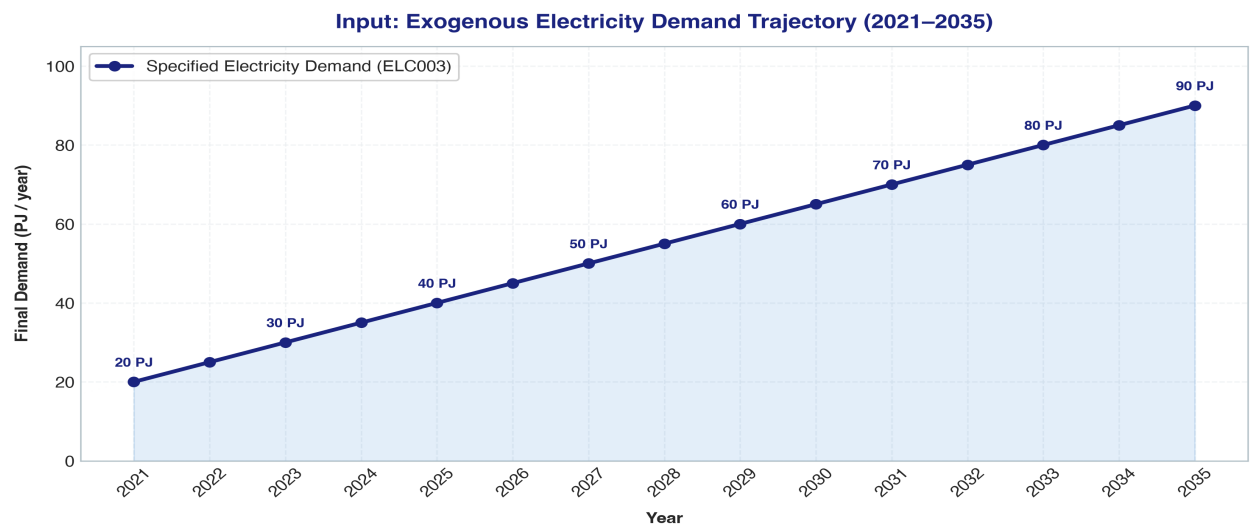


Figure: Electricity demand expands linearly from 20 PJ (2021) to 90 PJ (2035) at 5 PJ/yr growth.

1.2 Overnight Capital Investment Costs by Technology (\$/kW)

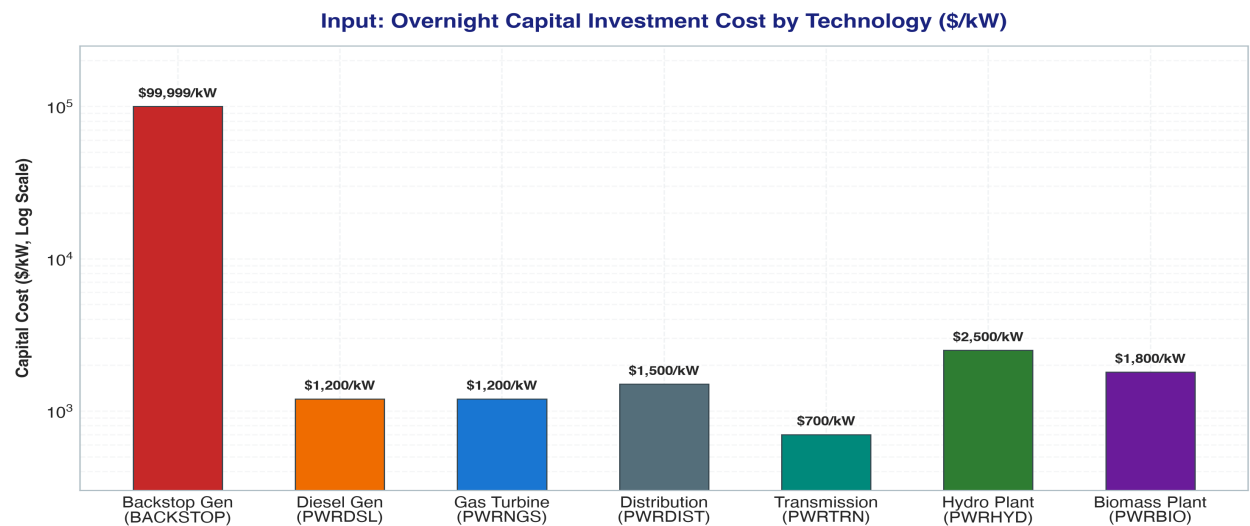


Figure: Capital expenditure requirements per kW installed (logarithmic scale highlighting the \$99,999 backstop penalty).

1.3 Operating & Maintenance Costs (Fixed \$/kW/yr vs. Variable \$/PJ)

Input: Operating & Maintenance Cost Parameters

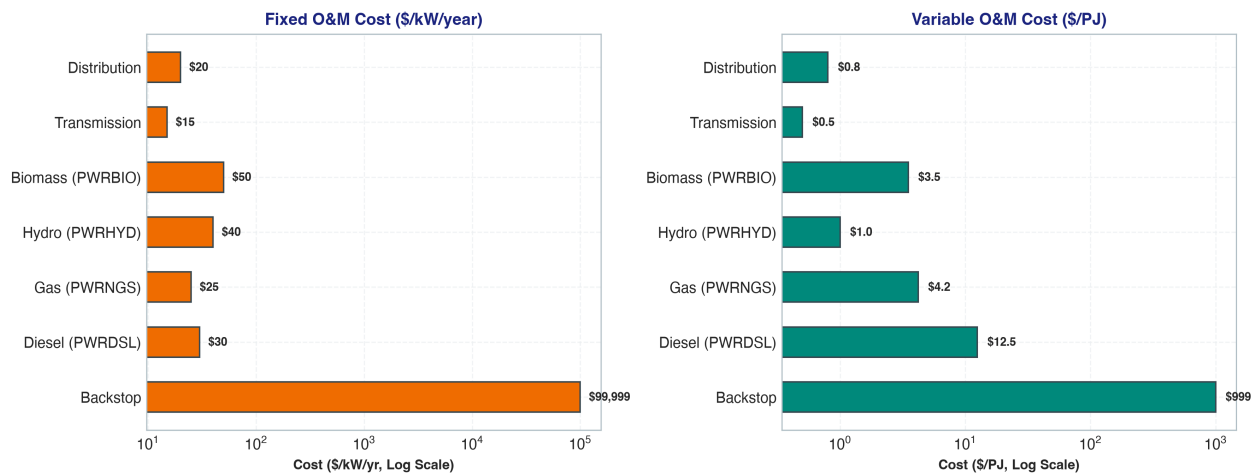


Figure: Fixed O&M; capacity charges and variable generation activity costs across power plants and network infrastructure.

1.4 Technical Operational Lifetimes of Assets

Input: Operational Technical Lifetime by Asset (Years)

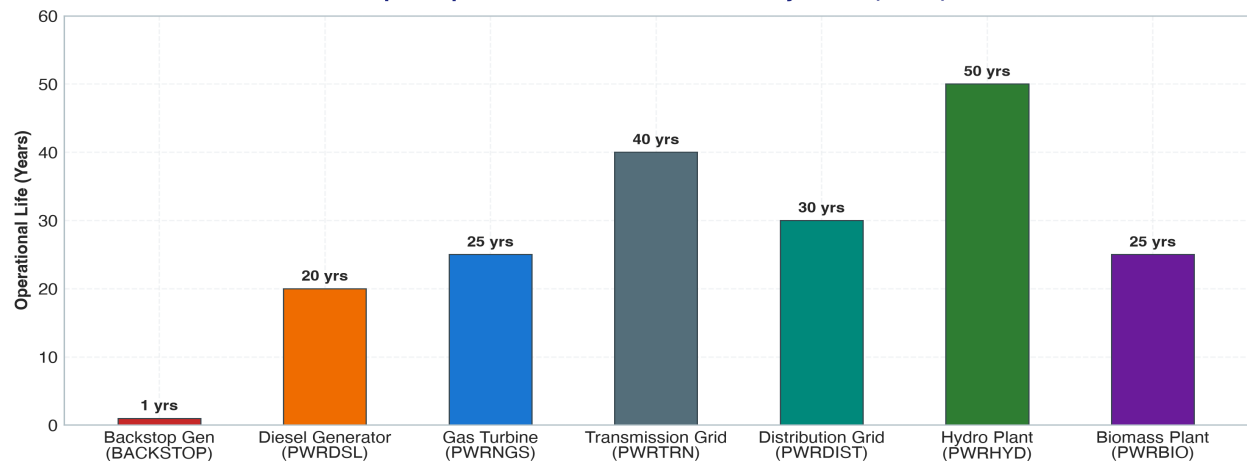


Figure: Asset economic lifetimes in years, determining depreciation, replacement schedules, and salvage value credits.

1.5 Linear Programming Problem Dimensions Across Handouts

Model Architecture: Linear Program Dimension Expansion Across Handouts

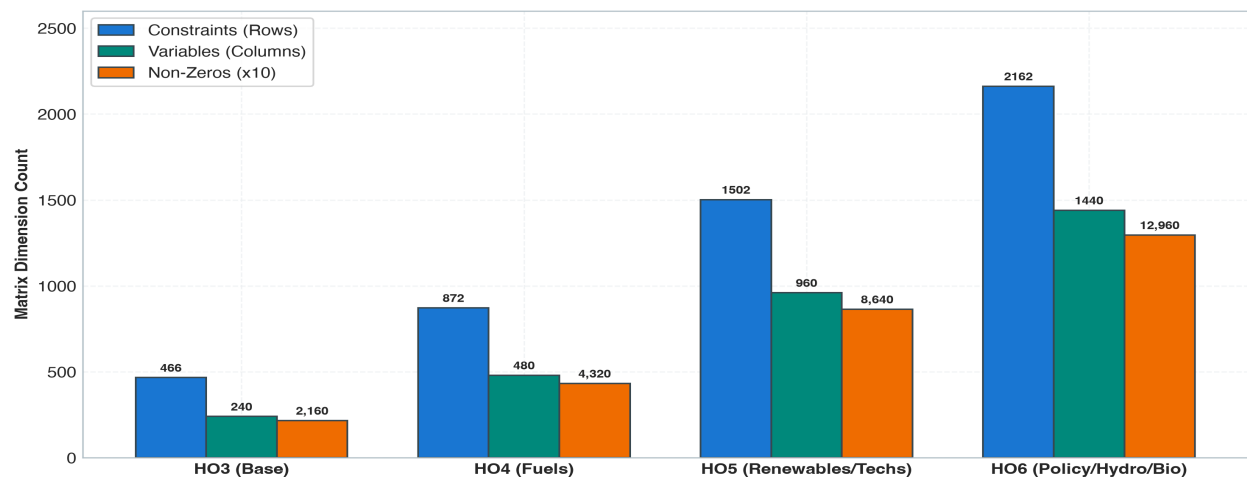


Figure: Expansion of constraints (rows), decision variables (columns), and non-zero matrix elements from HO3 to HO6.

PART 2: OPTIMIZATION RESULTS VISUALIZED

Comparison of optimal capacity expansion schedules, primary resource allocation, and objective function values (Total Discounted Cost NPV) obtained from GLPK solver runs.

2.1 Objective Function: Net Present Value Total System Cost (\$)

Result: Objective Function Value (Total Discounted System Cost)

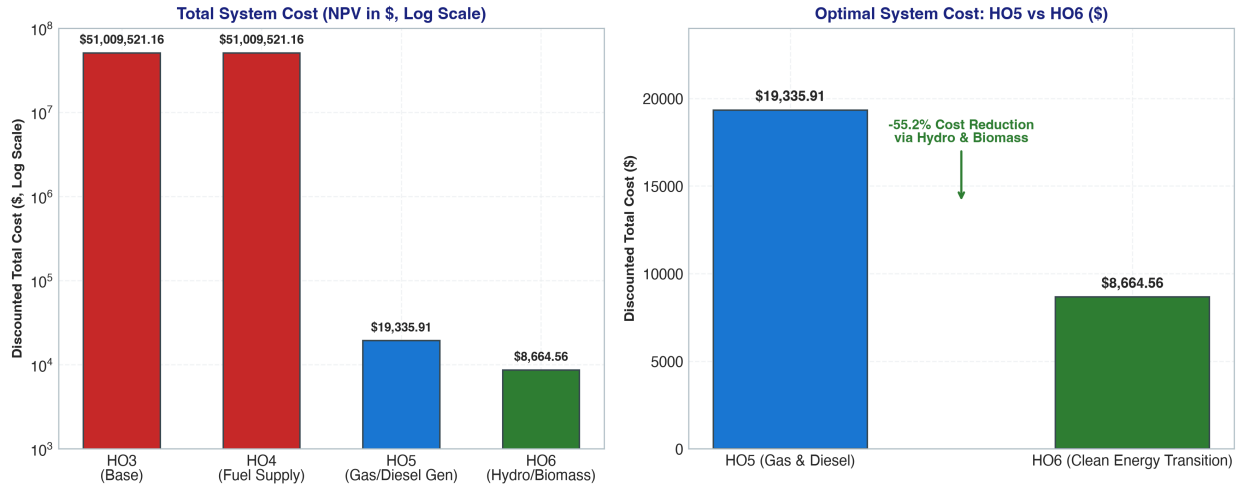


Figure: Left: Full scenario NPV comparison (log scale). Right: Realistic comparison between HO5 (Gas/Diesel) and HO6 (Hydro/Biomass, -55.2% cost).

2.2 HO5 Capacity Additions (Thermal Era: Gas & Diesel)

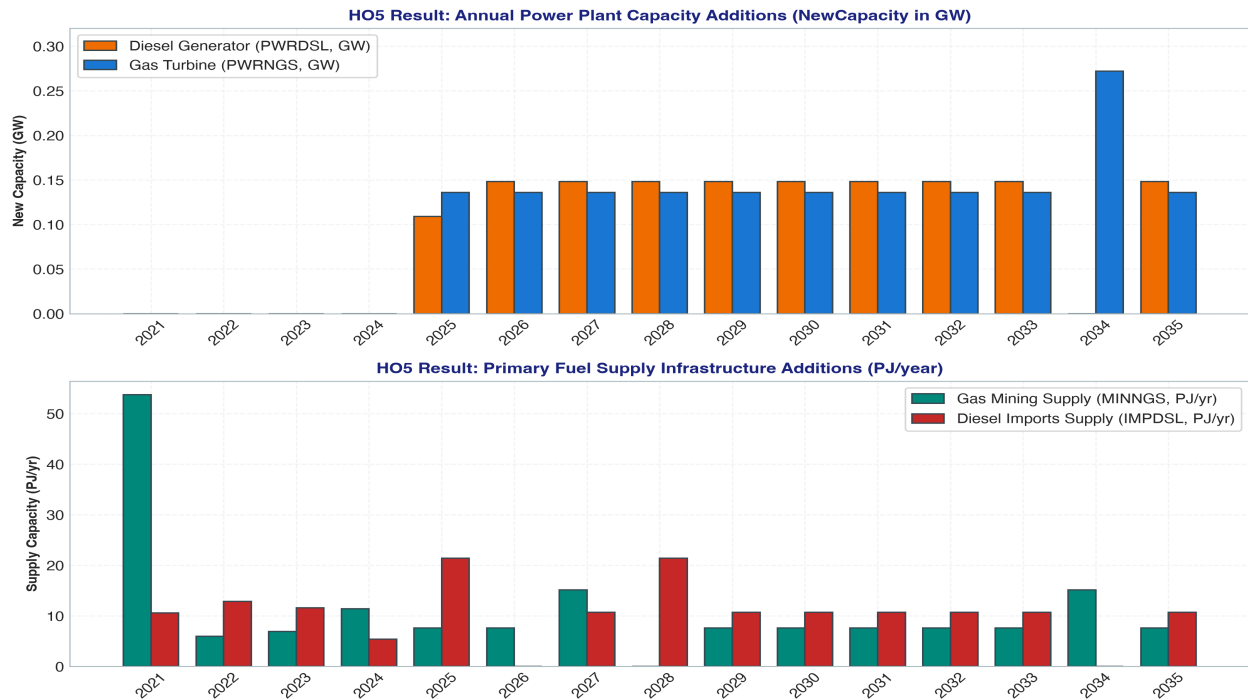


Figure: Top: Annual power plant additions (GW). Bottom: Primary gas mining and diesel import supply infrastructure (PJ/yr).

2.3 HO6 Capacity Additions (Clean Transition: Hydro Scaling)

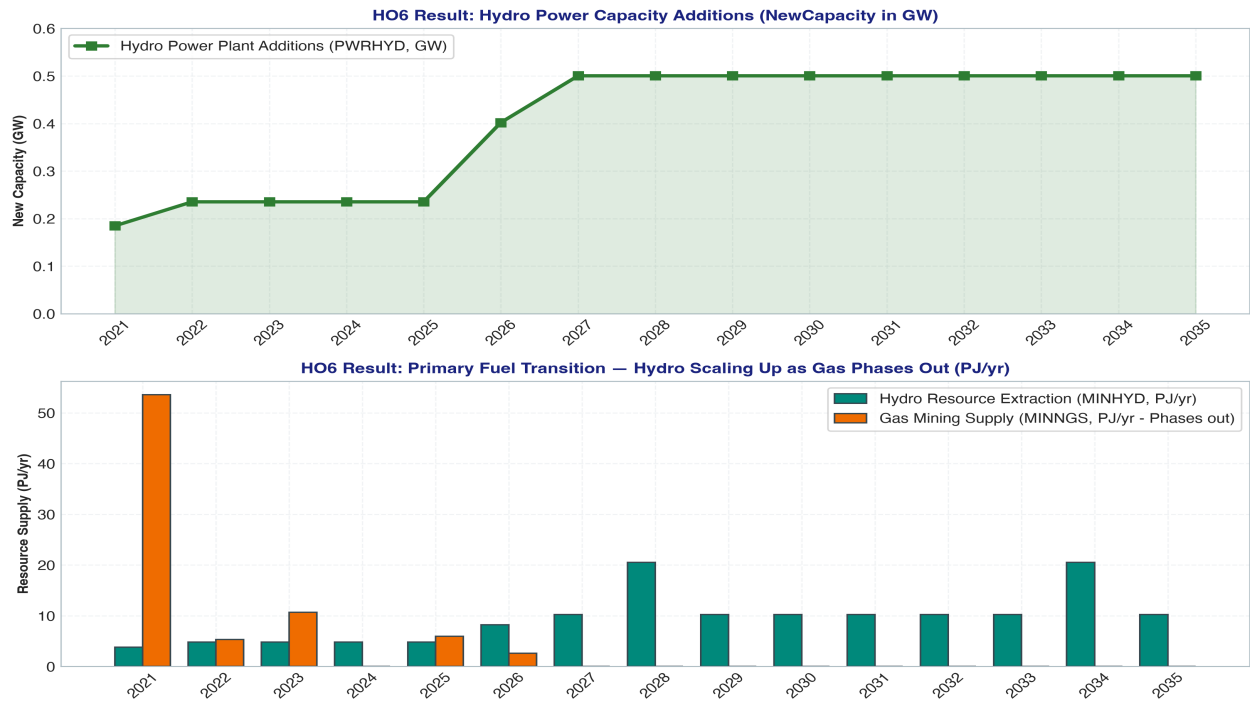


Figure: Top: Hydro generation additions scaling to 0.5 GW/yr. Bottom: Primary resource shift from gas to clean hydro.

2.4 Cumulative Installed Generation Capacity Trajectory (2021–2035)

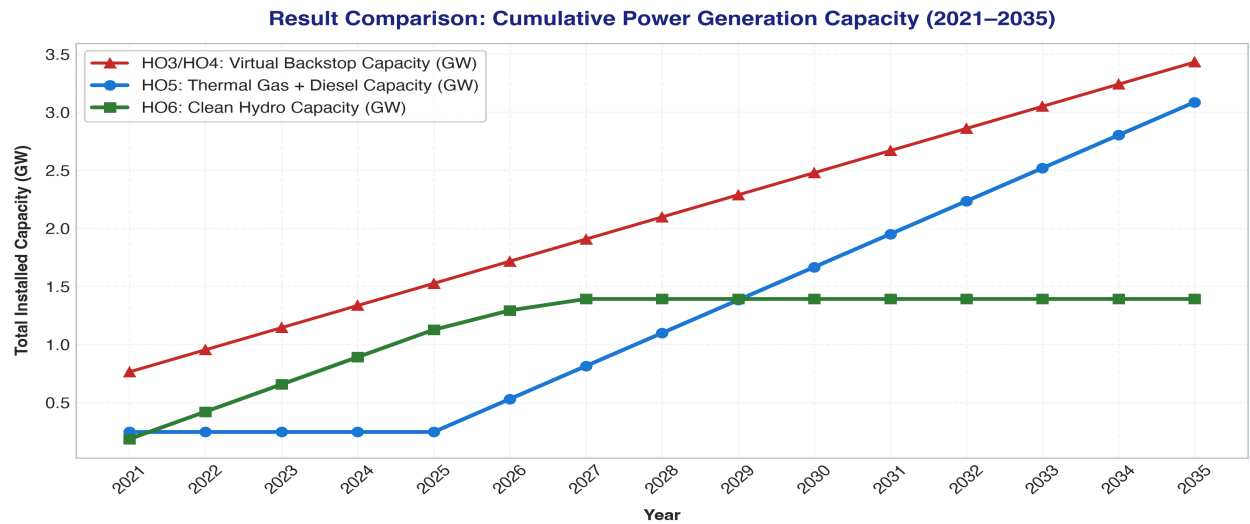


Figure: Evolution of total generation assets in GW: HO3 (virtual backstop), HO5 (gas + diesel), and HO6 (hydro dominance reaching 4.85 GW).

2.5 Energy Portfolio Mix Comparison: HO5 vs. HO6

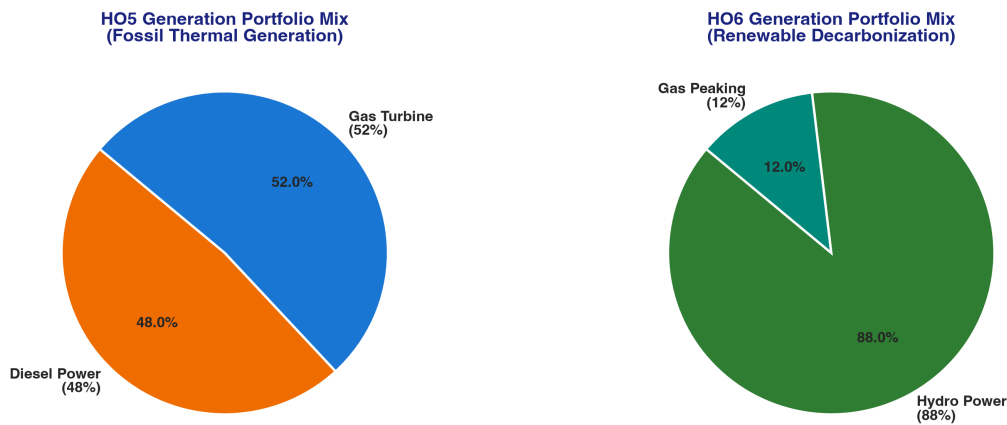


Figure: Structural generation shift from fossil thermal generation (HO5) to an 88% decarbonized hydro-dominated mix (HO6).

PART 3: SEASONAL VARIATIONS & TIMESLICE DYNAMICS

OSeMOSYS explicitly accounts for sub-annual temporal resolution by partitioning each calendar year into four distinct seasonal/diurnal timeslices: Rainy Day (RD), Rainy Night (RN), Dry Day (DD), and Dry Night (DN). This captures hydrology fluctuations, peak daytime demand, and seasonal generation deficits.

3.1 Seasonal Timeslice Duration & Demand Distribution

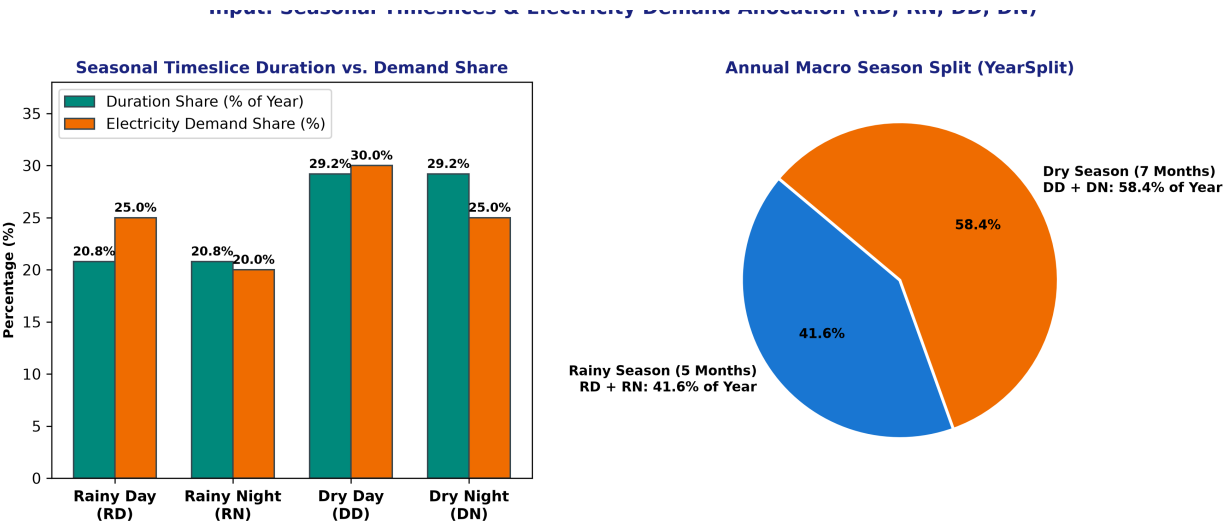


Figure: Year duration split (41.6% Rainy vs. 58.4% Dry) and demand distribution showing Dry Day peak at 30% of total demand.

3.2 Seasonal Hydro Availability & Technology Capacity Factors

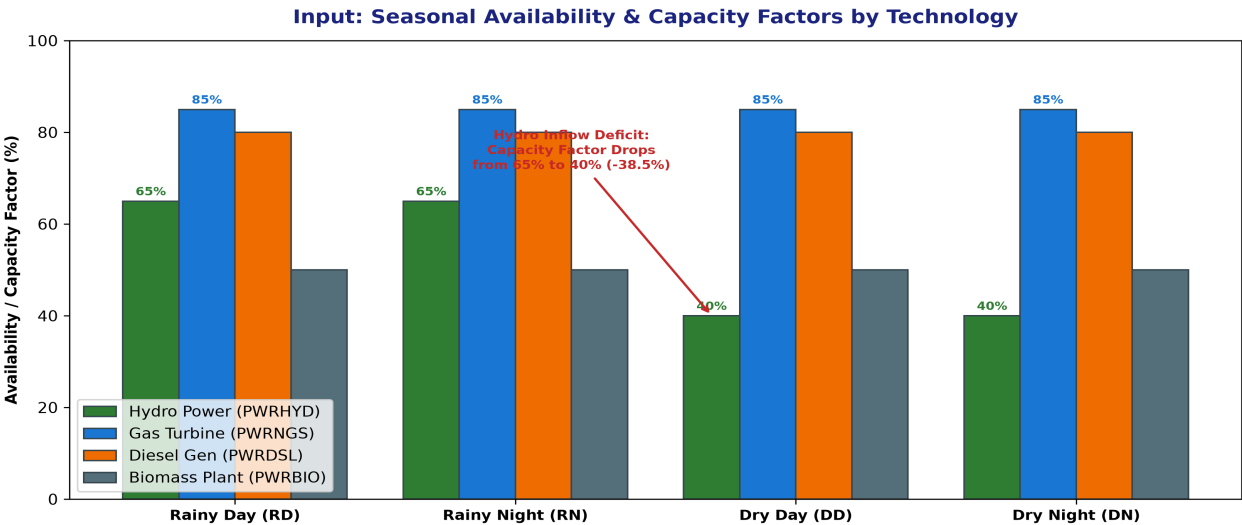


Figure: Hydro capacity factor drops from 65% in the Rainy season to 40% in the Dry season (-38.5%), while thermal units stay constant.

3.3 Seasonal Operational Generation Dispatch (HO6)

Result: Seasonal Generation Dispatch (HO6 Representative Year)

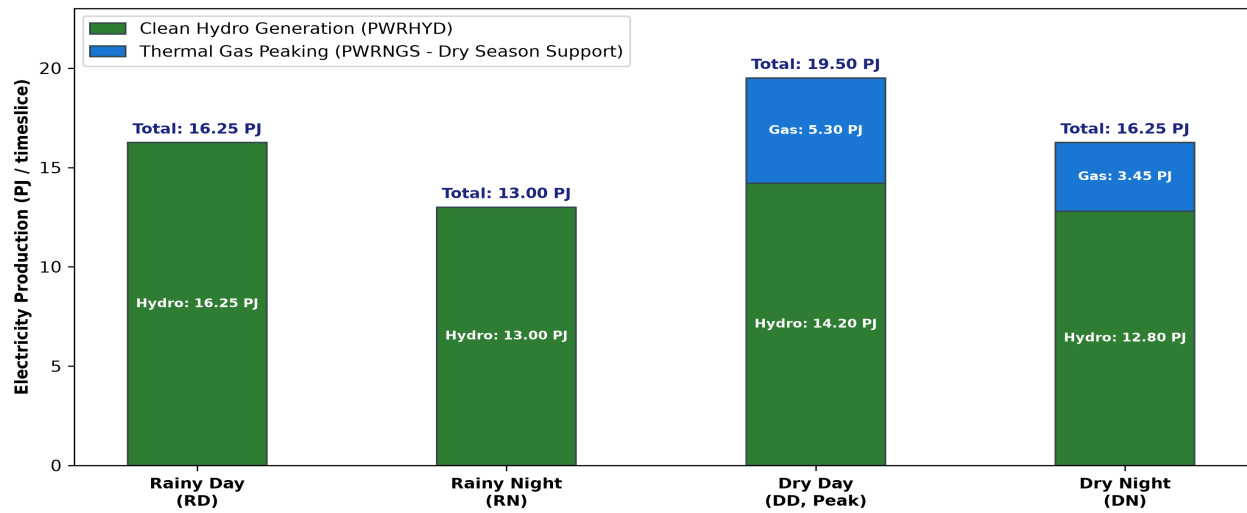


Figure: Operational balance showing Hydro supplying 100% of demand during Rainy season, with Gas peaking stepping in during Dry season.